



Geolocated social media as a rapid indicator of park visitation and equitable park access



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ABSTRACT

Understanding why some parks are used more regularly or intensely than others can inform ways in which urban parkland is developed and managed to meet the needs of a rapidly expanding urban population. Although geolocated social media (GSM) indicators have been used to examine park visitation rates, studies applying this approach are generally limited to flagship parks, national parks, or a small subset of urban parks. Here, we use geolocated Flickr and Twitter data to explore variation in use across New York City's 2143 diverse parks and model visitation based on spatially-explicit park characteristics and facilities, neighborhood-level accessibility features and neighborhood-level demographics. Findings indicate that social media activity in parks is positively correlated with proximity to public transportation and bike routes, as well as particular park characteristics such as water bodies, athletic facilities, and impervious surfaces, but negatively associated with green space and increased proportion of minority ethnicity and minority race in neighborhoods in which parks are located. Contrary to previous studies which describe park visitation as a form of nature-based recreation, our findings indicate that the kinds of green spaces present in many parks may not motivate visitation. From a social equity perspective, our findings may imply that parks in high-minority neighborhoods are not as accessible, do not accommodate as many visitors, and/or are of lower quality than those in low-minority neighborhoods. These implications are consistent with previous studies showing that minority populations disproportionately experience barriers to park access. In applying GSM data to questions of park access, we demonstrate a rapid, big data approach for providing information crucial for park management in a way that is less resource-intensive than field surveys.

1. Introduction

Understanding why some parks are used more regularly or intensely than others can inform ways in which urban parkland is developed and managed to meet the needs of a rapidly expanding urban population. Since 2008, the global population has been mostly urban. By 2050, urban populations are expected to make up 70–80% of the world's total population (United Nations, 2014). As cities expand and grow denser, inner-city parks will be increasingly important for improving environmental quality, human health, and neighborhood livability. In addition to the challenge of expanding service provision, local governments also

face challenges associated with urban patterns of social segregation, community disinvestment, and disproportionate access to public goods. Thus, a key sustainability challenge for local governments is to develop and maintain parks and other public goods in ways that equitably distribute benefits to health, well-being, livability, and the economy. Exploring inner-city park use and correlates of use can help bring to light ways in which these benefits are distributed.

Using field techniques such as on-site counts or surveys, visitor use information is often collected for national parks, large flagship parks and parks that draw large numbers of tourists. However, our knowledge of a broader range of urban parks – particularly small and medium-

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sized community parks which may be integral to the daily routines of urban residents – is limited. Geolocated social media (GSM) is increasingly being used to gain insight into human behavior patterns – from social movements like Arab Spring and Occupy Wall Street to retail consumption patterns in tanning salons and flea markets (Quercia & Saez, 2014). Although GSM indicators have also been used to examine park visitation, studies applying this approach are generally limited to flagship parks, national parks, or a small subset of urban parks (e.g., Hausmann et al., 2017; Schwartz & Hochman, 2014; Sessions, Wood, Rabotyagov, & Fisher, 2016; Wood, Guerry, Silver, & Lacayo, 2013). Here, we use geolocated Flickr and Twitter postings as data to explore variation in use across New York City's 2143 diverse parks, and model visitation based on spatially-explicit park characteristics and facilities, neighborhood-level accessibility features and neighborhood-level demographics. We demonstrate a GSM approach for providing indicators of park visitation in a way that is less resource-intensive than field surveys.

2. Literature review

2.1. Park visitation and its determinants

Trends in park visitation along geographic, temporal, and socio-cultural dimensions can inform ways in which public agencies and other entities spatially plan, design and manage these resources. Regular visitation estimates are produced for park units in the United States National Park Service system using approaches such as adjusted traffic counts or temporally limited visual pedestrian counts extrapolated over longer time periods (Bruilliard, 2016). However, most studies exploring urban park visitation and drivers of urban park use have limited the number of parks examined to subsets selected according to particular criteria, including most frequently visited parks (Buchel & Frantzeskaki, 2015), importance to particular communities and stratified by socio-economic status (Cohen et al., 2013), stratified by neighborhood (Kaczynski, Potwarka, & Saelens, 2008), or a random sample (Baran et al., 2014). Traditional methods rely on observation and survey questionnaires to ascertain visitation for particular parks or variation in park use over geography, time, socio-cultural groups or activities performed by park users (e.g., Baran et al., 2012). For decades, researchers have argued that park use estimates inferred from limited visual observations could be combined with approaches such as visual, traffic and mechanical counts to help automate estimation procedures (Mowen, 2002). In New York City (NYC), Silver, Giorgio, and Mijanovich (2014) examined the use of ten small parks and playgrounds by conducting a series of interviews with adult park users, which typifies the methodological approach to studying park use in NYC. To date, there is not a single comprehensive study that examines park use across all NYC parks. Here, we suggest that GSM data can be used to supplement existing temporally and spatially-limited estimation procedures as an automated and less resource-intensive way of revealing park use variation and determinants.

Determinants of park use that have been identified through a broad range of leisure studies, public health, as well as child and youth studies include park characteristics such as size, recreational facilities, physical activity facilities (Baran et al., 2014; Kaczynski et al., 2008), programming (Tester & Baker, 2009), neighborhood conditions such as safety, availability of pedestrian access facilities, urban form, socio-economic conditions, racial heterogeneity (Baran et al., 2014; Van Dyck et al., 2013; Gomez, Baur, Hill, & Georgiev, 2015) and socio-cultural characteristics of park users (Baran et al., 2012; Mowen, Payne, & Scott, 2005; Lin, Fuller, Bush, Gaston, & Shanahan, 2014). Conversely, determinants of park nonuse may include financial constraints, time constraints, long travel distances, inadequate park space, racial discrimination, inequitable park programming, and park users' perceptions that parks are unwelcoming or unsafe (Byrne, 2012). Current and historic social relations, park design and park management can play key

roles in mediating these determinants. Exclusionary practices on the basis of race and ethnicity (Cutts, Darby, Boone, & Brewis, 2009), income (Crawford et al., 2008), age, language and other social characteristics can be embodied in park design (Byrne, 2012), public space policies that deter activities preferred by social subsets (Owens, 2002) or practices that effectively ostracize and discriminate (Byrne, 2012). These observations are not limited to urban parks that are solely managed by local governments. A comprehensive survey of the National Park System in the United States that was conducted in 2000 – which supported an earlier study conducted from 1982 to 1983 – indicated that smaller proportions of racial and ethnic minorities tended to visit national park system units relative to Whites (Floyd, 1999; Solor, Hagen, & Ostergren, 2003). Hispanic Americans were more likely to be more concerned about safety than non-Hispanics, and African Americans were more likely than Whites to feel uncomfortable in parks and believe that park employees provided poor levels of service. Given that public park resources are often inequitably distributed, a key concern for public administrators is to understand variation in those resources across geography, socio-economic and demographic conditions, toward developing programming that more evenly distributes benefits.

2.2. Geolocated social media (GSM) data indicators of visitation and visitation drivers

Big data produced through social media networks – particularly micro-blogging platforms such as Twitter and Flickr – have recently given rise to new branches of social and geographic research and the inception of so-called digital social sciences (Edwards, Housley, Williams, Sloan, & Williams, 2013; Housley et al., 2014) and a “neo-geography” (Antoniou, Morley, & Haklay, 2010) based on user-generated spatial content. Such new data and methods create potential to augment conventional approaches for studying urban parks, drivers of their use and nonuse as well as the social and public health benefits they provide. While the use of GSM data in urban parks research is still in its infancy, over the past few years, a growing number of studies have used GSM data for research in the fields of geography (Lin & Cromley, 2015; Naaman, 2011; Prager & Wiegand, 2014), landscape and urban planning (Dunkel, 2015; Shelton, Poorthuis, & Zook, 2015; Sonter, Watson, Wood, & Ricketts, 2016), and urban ecology (Guerrero, Møller, Olafsson, & Snizek, 2016). Scholars have emphasized new opportunities that data at this scale with nearly real-time monitoring provides for research and practice across multiple scientific domains. In urban parks research, the multiplicity of digital traces that people leave – from tweets, to images, text, tags, and shared locations – can provide new insights into which public investments (e.g., urban parks) are most used, how people are using them and why. We suggest that such knowledge could be an important new source for informing public policy and supporting citywide planning efforts.

Over the past few years, a subset of GSM research has suggested that geotagged social media data can serve as a proxy for visitation data at a variety of public venues (including parks) (Heikinheimo et al., 2017; Keeler et al., 2015; Sonter et al., 2016; Wood et al., 2013) and aid the discovery of major points of interest in the urban environment. Social hubs, routine activity patterns, hotspots of social activity and tourism, as well as work-related activities have been studied using social media in New York, Paris, London, Los Angeles, Chicago, and elsewhere (Bawa-cavia, 2011; Ferrari, Rosi, Mamei, & Zambonelli, 2011; Girardin, Fiore, Ratti, & Blat, 2008; Jankowski, Andrienko, Andrienko, & Kisilevich, 2010; Martí, Serrano-Estrada, & Nolasco-Cirugeda, 2017; McKenzie, Janowicz, Gao, & Gong, 2015; Steiger, Westerholt, Resch, & Zipf, 2015).

Studies in the U.S. and Europe have used GSM to predict and model visitation in national parks and conservation areas. More specifically, geotagged Flickr images have been found to be a reliable proxy for the visitation of lakes in Minnesota and Iowa (Keeler et al., 2015), over 30

national parks in western U.S. (Sessions et al., 2016), and > 800 recreational sites around the world (Wood et al., 2013). With respect to urban parks, GSM data have primarily been used to study internal visitation patterns rather than test data validity as a proxy for visitation. Geotagged photographs from Instagram have, for instance, been used to characterize the distribution of people in three urban parks in New York City – Madison Square Park, Bryant Park, and Union Square (Schwartz & Hochman, 2014). However, GSM data may also offer ways for park managers to understand the entire distribution of visitors across all parks within a single city.

In addition to simply measuring visitation, GSM data are increasingly being used to identify predictors of visitation. Characteristics such as spatial configuration, landscape features, and amenities determine GSM-based visitation patterns. In urban parks, features such as art installations, food venues, skating rinks, public programming, landmark buildings (Schwartz & Hochman, 2014) and recreation activities (Cord, Roeßiger, & Schwarz, 2015) influence visitation. In regional parks, water quality (Keeler et al., 2015), Red-List species (Willemen, Cottam, Drakou, & Burgess, 2015) and landscape features (Sonter et al., 2016) have been identified as important predictors of use. Who uses parks – whether tourists or visitors (García-Palomares, Gutiérrez, & Mínguez, 2015; Girardin et al., 2008) – and how users vary depending on park amenities, landscape features and environmental quality (Dunkel, 2015; Keeler et al., 2015; Sonter et al., 2016) can shed light on areas where future visitation may tend to concentrate and what can be done to rectify current asymmetries in use and access.

3. Methods and data

3.1. Study area

NYC is the largest and most dense city in the USA with roughly 8.4 million inhabitants (U.S. Census Bureau, 2015), and an average population density of 10,429 inhabitants per square kilometer. Twenty percent of NYC's land area is parkland which makes it one of the greenest cities in the U.S. However, despite a large amount of park space, many New Yorkers do not have easy and local access to park benefits. For example, < 40% of the population is within a 10-minute walk of a park (Trust for Public Land, 2017). To address this, PlaNYC (New York City, 2011) – one of the City's recent sustainability plans – established a goal for every resident to be within a 10-minute walk of a park, while the most recent OneNYC plan focuses on improving park quality through facilities and amenities upgrades (nyc.gov/onenyc). While NYC agencies are making concerted efforts to increase green infrastructure (Grove et al., 2006; NYC Environmental Protection, 2010), population growth continues to place pressure on the city to provide parks and park amenities to underserved populations.

NYC parks are primarily managed by the Department of Parks & Recreation (NYC DPR). Many local non-profit agencies – such as the Central Park Conservancy (CPC) and Prospect Park Alliance – provide additional management for specific parks and areas of interest through public-private partnerships. However, many parks do not have such institutions and are in disrepair. Since many of the city's parks and public spaces were designed 50 to 100 years ago, significant investment is needed to upgrade and meet shifting demands and goals (OneNYC).

Survey-derived park visitation information exists for some flagship parks, though surveys are still infrequent and often have low sample size. In 2008 and 2009, the CPC conducted the first systematic effort since 1873 to measure an entire year of Central Park's public use (Central Park Conservancy, 2011). While this effort is valuable for managing Central Park, these data are rarely collected for other parks in the city. With the most ethnically and linguistically diverse urban population in the U.S., effective targeting of future park investments in a way that equitably meets the needs of all NYC residents requires understanding what drives variability in parks usage.

Our objective is to describe the spatial variation of GSM-based

visitation to parks in NYC, explore potential drivers of visitation variation – including park characteristics, accessibility to parks and neighborhood social characteristics – and to explore whether drivers vary by resident or tourist visitation. We also aim to validate the use of GSM data as an indicator of broader visitation trends for a subset of parks for which survey-based visitation data are available.

3.2. Park visitation & construction of dependent variables

We constructed park usage rate estimates called Flickr user days (FUD) and Twitter user days (TUD) following methods described by Wood et al. (2013) at the park-level for all 2143 parks in NYC based on a GIS shapefile provided by the NYC DPR (2013). Flickr's Application Programming Interface (API) (Flickr, 2015) provides access to information on over 250 million publicly geotagged photographs taken between 2005 and 2014. Locations and dates associated with photographs refer to when and where the image was taken (it may have been uploaded to Flickr at a later date). For each park, we spatially queried this global set of images and counted the unique combinations of Flickr usernames and dates per park to compute the total FUD each year over this time period. Data from Twitter was provided by the Twitter Streaming API (Twitter, 2015) that releases a random 1% of all public tweets, which amounted to 51.3 million geotagged tweets posted between 2012 and 2014 from within a rectangle bounding the study area. For each park, we calculate TUD in the same way as FUD, as the unique combination of users and dates. Average annual FUD and TUD were also calculated.

Since our aim is to understand correlates of visitation not only to flagship parks which draw large numbers of tourists, but also to smaller community parks that are regularly visited by local residents (or could provide important benefits to local residents), we divided the Flickr user population into residents and tourists. We determined the home town of individuals who shared photographs of NYC parks on Flickr based on the self-reported locations shared on their public Flickr profile (sensu Wood et al., 2013). Out of 54,330 unique Flickr users accounting for the FUDs across the study area, 24,458 reported their home location, and over 96% of those locations were successfully geocoded to latitude/longitude coordinates using Twofishes (www.twofishes.net). We designated users as 'residents' (24.4%) if their geocoded home location was within a 1-mile buffer of the NYC boundary and 'tourists' (75.6%) if not. Then we constructed yearly mean resident FUD and tourist FUD for each park by filtering the Flickr photo database by resident and tourist users respectively, prior to calculating user-days. We did not divide TUD into residents versus tourists because equivalent data on hometowns were not provided by Twitter users.

3.3. Predictors of park visitation rates

We tested predictors of park visitation in three categories: 1) park facilities and characteristics, 2) park accessibility and 3) neighborhood characteristics (Table 1). For park facilities and characteristics, we constructed predictor variables that may draw park visitors or accommodate larger numbers of visitors. These include area of the park, green space, impervious surface, water bodies such as lakes or beaches, sports facilities, wifi hotspots, play areas, and whether the park is listed as a playground, flagship park community park, or nature area. Larger parks may have a greater capacity to accommodate more people and draw from a larger catchment area. We also hypothesized that green spaces, water bodies sports facilities, wifi hot spots and play areas would attract people who are interested in nature-based recreation, sports-based recreation and other facilities. We expected that local residents would be drawn to playgrounds, as these can provide important facilities for children and families who may otherwise have limited access to open spaces for play. Due to their national and international reputations, we expected that flagship parks such as Central Park and Prospect Park would have relatively high visitation rates – particularly by tourists –

Table 1
Park and neighborhood-level predictor variables, metrics used to measure each, spatial scale of measurement and data source.

Predictor	Metric	Spatial scale	Source
Park facilities and characteristics	Area	Park-level	NYC DPR
	Green space	Park-level	(MacFaden, O'Neil-Dunne, Royar, Lu, & Rundle, 2012)
	Impervious surface	Park-level	(MacFaden et al., 2012)
	Water	Park-level	(MacFaden et al., 2012)
		Within 0.1 mile of park	(MacFaden et al., 2012)
		Park-level	(MacFaden et al., 2012)
		Park-level	
	Sports facilities	Park-level	NYC DPR
		Park-level	NYC DPR
		Park-level	NYC DPR
Neighborhood socio-demographic characteristics	Wifi	Park-level	NYC DoITT
	Play areas	Park-level	NYC DPR
	Park type	Park-level	NYC DPR
		Park-level	NYC DPR
		Park-level	NYC DPR
		Park-level	NYC DPR
		Park-level	NYC DPR
		Park-level	NYC DPR
		Within 400 m of park	Census, 2010
		Within 400 m of park	Census, 2010
Accessibility	Race	Within 400 m of park	Census, 2010
	Ethnicity	Within 400 m of park	Census, 2010
	Poverty	Within 400 m of park	Census, 2010
	Income	Within 400 m of park	ACS, 2013
	Vacancy	Within 400 m of park	Census, 2010
	Property value	Within 400 m of park	Census, 2010
	Bicycle	Within 400 m of park	NYC DOT
		Within 400 m of park	NYC DOT
		Within 400 m of park	NYC DOT
		Within 400 m of park	NYC DOT
Large roads Crime	Distance to nearest bus stop (100 m)	Within 400 m of park	NYC DOT
	Subway stops (count)	Within 400 m of park	NYC DOT
	Distance to nearest subway stop (100 m)	Within 400 m of park	NYC DOT
	Bus and subway stops (count)	Within 400 m of park	NYC DOT
	Interstate and state-recognized roads (count)	Within 400 m of park	NYC DOT
	Yearly average crime (count/1000 people)	Within 400 m of park	Census Tiger line
		Within 400 m of park	SpotCrime, 2007–2010

Sources abbreviations include the following:

NYC DPR: New York City Department of Parks and Recreation.

NYC DoITT: New York City Department of Information Technology and Telecommunications.

NYC DOT: New York City Department of Transportation.

ACS: American Community Survey.

and that community parks would be relatively popular with local residents. We also expected that parks classified as Nature Areas by the NYC Parks Department would draw relatively high numbers of visitors, as they provide natural resources that are not easily accessible for many New Yorkers. All park facilities and park characteristics predictor variables were created at the park or near park-level, as specified in Table 1.

At the neighborhood scale, we measured transportation access and socio-demographic characteristics. We hypothesized that parks in closer proximity to a larger number and extent of public transportation facilities – including bicycle, bus, and subway infrastructure – will be more frequently visited. By contrast, we expected that parks in close proximity to large roads which may cut off access or generate noise and air pollution will receive fewer visitors. Neighborhood socio-demographic characteristics – including population density, proportions of minority races and ethnicities, people living below the poverty line, vacant units, property values and crime – were also evaluated. By virtue of proximity to large numbers of people, we expected that parks located in communities with relatively high residential population densities would also exhibit relatively high visitation rates. Due to ways in which historic cycles of poverty and disinvestment brought about by redlining and other policies related to race and class are often manifested in public resources such as parks and other community infrastructure, we expected that parks located in neighborhoods with relatively high proportions of minority races, minority ethnicities, people living below the poverty line, vacant units, crime, low household income and low property values would have relatively low GSM-based visitation rates. Parks located in neighborhoods with large proportions of low socioeconomic status residents may have poorer quality facilities and may be less likely to be visited by tourists due to poor accessibility.

We defined the neighborhood of each park as a 400m doughnut surrounding the park, and computed all neighborhood characteristic variables using a Euclidean buffer procedure. Many different approaches could be used to define a neighborhood; we employed a procedure based on NYC's definition of accessibility to parks as a 10-minute walk, a distance which is also commonly used in accessibility literature (e.g., Miyake, Maroko, Grady, Maantay, & Arno, 2010). Population, race, ethnicity, poverty, vacant housing, property value and income data were all acquired from the U.S. Census. The finest spatial unit at which U.S. Census variables used in this study are available is the block group-level. While this spatial unit captures variation in demographic information across cities like NYC, it can be problematic for measuring access, particularly when using a buffer procedure. Block groups contain all residential and non-residential areas, including industrial or manufacturing areas, schools and other institutions, parking lots and parks. Depending on the size of a block group and the distribution of land uses and residential populations within that spatial unit, using average populations figures at the block group-level can be problematic. For instance, Central Park is approximately 3.4 km² and the block group that contains Central Park is slightly larger – approximately 3.6 km²; the boundary of the Census block group contains the inner boundary of the doughnut buffer around Central Park. Thus, when computing population values within the Central Park buffer, the block group containing the park itself will be included (Fig. 1). This is highly problematic, as the large area of the park will substantially lower the estimated population density value. The arbitrary nature of Census block group units, combined with a lack of coordination among agencies that create the multiple datasets being used in this analysis presented a challenge for estimating population values in a way that reflects their true distributions.

In order to manage this problem, we applied a dasymetric mapping technique using an ArcGIS tool developed by the USGS (Sleeter & Gould, 2007). Dasymetric mapping is a disaggregation technique which uses an ancillary dataset representing the population statistical surface to redistribute total population values according to a set of empirically-derived weights. It is particularly useful for applications to

environmental and social justice, in which accurate estimations of people who have access to a resource or are exposed to risk is crucial (Mennis, 2002). We used the land use field of the NYC tax lot dataset (NYC Department of City Planning, 2011), which contains four residential classes: 1) single family/duplex, 2) multi-family elevator, 3) multi-family walk-up, 4) mixed use residential and commercial. We rasterized this tax lot dataset to a cell size of 5 m and reclassified non-residential land use classes as “0.” Using the block group-level Census datasets, we sampled population values in each of the four residential classes and redistributed population values at the 5m spatial scale. The final dasymetric maps better reflect residential land uses, more accurately illustrate population distributions in NYC, and enable us to capture park access to neighboring communities in a more precise way than would Census block group units.

For instance, computing the population density within a 400 m buffer of Central Park using the Census block group spatial unit yields a population figure of 24.1 people/1000 m², whereas the estimated density using the dasymetric 5m cellular unit is roughly a third greater – 36.4 people/1000m². Indeed, the residential area around Central Park contains many high-rise multi-family buildings, and provides park access to a large number of the city's residents that are undercounted at the aggregate Census block group unit.

We used the dasymetric disaggregation approach to construct all neighborhood characteristic variables for which a total population value is available – including total population, White population, African American or Black population, Hispanic or Latino population, population living below the poverty line, and vacant units. We then transformed total population to a density, and all other values as the percent of total population or residential units (in the case of vacant units). We also constructed an average median property value variable, and yearly average crime as the number of crimes per 1000 people within each buffer.

We constructed all neighborhood accessibility variables at the neighborhood-level, within 400 m buffers of each park, to match the population datasets. The 400m buffer doughnut is considered the area in which people are able to walk within 10 min (Miyake et al., 2010). We computed length of bike routes, and distance to the nearest bike route, bus stop and subway stops variables per 100 m using a near distance tool. We also constructed count variables for the number of bus stops, subway stops, total public transit stops (bus and subway) and interstate and state-recognized roads using a spatial join with buffer distance procedure.

3.4. Regression modeling

To understand whether park facilities and characteristics, neighborhood characteristics and accessibility measures influence visitation rates, we used simple linear regression to test which predictor variables (Table 1) explain FUD for all visitors, residents and tourists and TUD use estimates as dependent variables. Because many parks had zero or small FUD and TUD values (Appendix A), we also used logistic regression models to determine whether the presence or absence of these social media-based metrics is influenced by different variables than the variation in visitation among the parks with non-zero visitation rates. For parks with non-zero FUD and TUD rates, we used a negative binomial model distribution. Negative binomial regression is appropriate for over-dispersed count variables in which the conditional variance exceeds the conditional mean. High and significant Chi² values for the likelihood ratio test that alpha equals zero indicate that for the non-zero FUD and TUD outcome variables, the negative binomial distribution was a more appropriate model than the standard poisson distribution (Tables 3 & 4). We used simple pairwise regression analysis and correlation matrices to eliminate non-significantly associated variables and potentially confounding variables prior to the regression modeling. We then used backwards stepwise regression on the remaining independent variables to reach the full model in which all independent variables



Fig. 1. Alignment of Census block group, park and park buffer boundaries. The block group boundary in which Central Park is contained includes residential areas bordering the park. When population density within the 400m buffer is averaged over Census block groups, density of the residential areas bordering the park are underestimated due to the inclusion of the park's area.

were statistically significant.

3.5. Validation

To assess the accuracy of social media-derived visitation rate estimates, we compared estimates from Flickr and Twitter to counts of observed visitors in nature areas of parks in NYC. Observations were conducted by the US Forest Service across 211 distinct areas within 40 unique parks during the summers (June–September) of 2013 and 2014 (Auyeung, Campbell, Johnson, Sonti, & Svendsen, 2016). We analyzed 33 parks, excluding those that had < 3 daily observations or which lacked accompanying geospatial data representing the boundary of the survey areas.

We aggregated counts to the park-level by summing across all areas and then calculating the average daily count for each park, based on the number of observation days for each park. We compared these against average daily FUD and average daily TUD using scatterplots and Pearson's correlation coefficients (Fig. 3). These Flickr and Twitter daily estimates were calculated from the same datasets described in Section 2.2, but limited to all photos and tweets taken within the park boundary between June 1 and September 30 of all available years.

4. Results

4.1. Distributions of social media visitation

The frequency distribution of FUD and TUD social media visitation rates are both positively skewed (Appendix A). The majority (95%) of FUD values fall between 0 and 13 user days, but seven parks have rates between 537 and 894 user days. The majority (95%) of TUD values fall between 0 and 503 user days, but six parks have rates as high as 7291–50,384 user days. Geographically, relatively high social media visitation rates are visible in parks along the south Brooklyn coastline, throughout Manhattan and north-central Queens, as well as Prospect Park, Brooklyn (Fig. 2). Out of 2143 parks, 901 had FUD values of zero and 134 had TUD values of zero.

For both FUD and TUD, Central Park, Bryant Park, Union Square Park, Flushing Meadows Corona Park, Prospect Park and Washington square Park are among the 10 most highly-visited parks. The majority of highly-visited parks are in the borough of Manhattan and none of the

10 most highly-visited parks are located in the borough of Staten Island (Table 2).

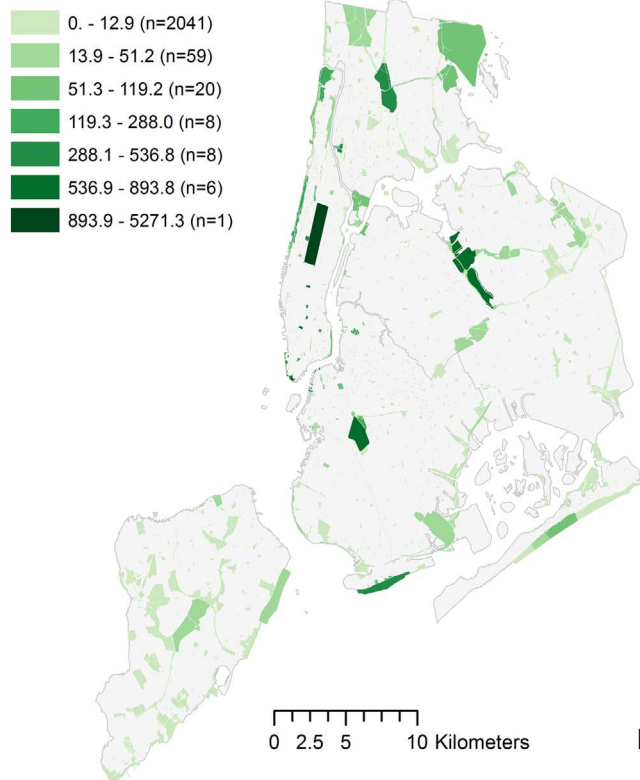
4.2. Analyses of Flickr photo user days and Twitter user days

Of the 35 explanatory variables representing park facilities and characteristics, neighborhood socio-demographic characteristics and accessibility (Table 1), 16 were selected by the backward stepwise regression as predicting variation in average annual FUD and 19 were related to average annual TUD (Table 3). The two models agreed on similar sets of predictor variables from the three categories; positive predictors for both models include area of park, number of water bodies in park, proximity to water bodies, the presence of a wifi hotspot, community park, length of bike routes, number of subway stops and proximity to nearest bike route, bus stop and subway stop. Negative predictors include area of green space, the park type being a playground and neighborhood-level minority race. Additional predictors of variation in non-zero FUD include play area (+), neighborhood-level minority ethnicity (–) and proximity to large roads (+), while additional predictors of variation in nonzero TUD include hard courts (+), nature area park type (+), neighborhood-level vacant units (+), neighborhood-level median property value (+) and proximity to large roads (–). Features such as area, number of water bodies and proximity to the public transportation types that we considered had generally low effect sizes for the units at which we measured them (100m² for areas and 100 m for distances), while the presence of a wifi hotspot, playground park type, community park type, vacant units, minority race and ethnicity and number of subway stops had relatively large effects (dummy variable, 0 vs 1). For instance, the presence of a wifi hotspot increases the log of expected FUD and TUD counts by 1.220 and 1.229 units, respectively, relative to parks that are not wifi hotspots. Each additional percentage of minority race in the neighborhood decreases the log of expected FUD and TUD by 1.821 and 0.864, respectively. Each additional subway stop in the neighborhood of a park increases the log of expected FUD and TUD by 0.447 and 0.425.

4.3. Analyses of resident and tourist Flickr photo user days and Twitter user days

Comparisons of resident and tourist models indicate that high social

Flickr photo user days (yearly mean) 2005–2014



Twitter user days (yearly mean) 2012–2014

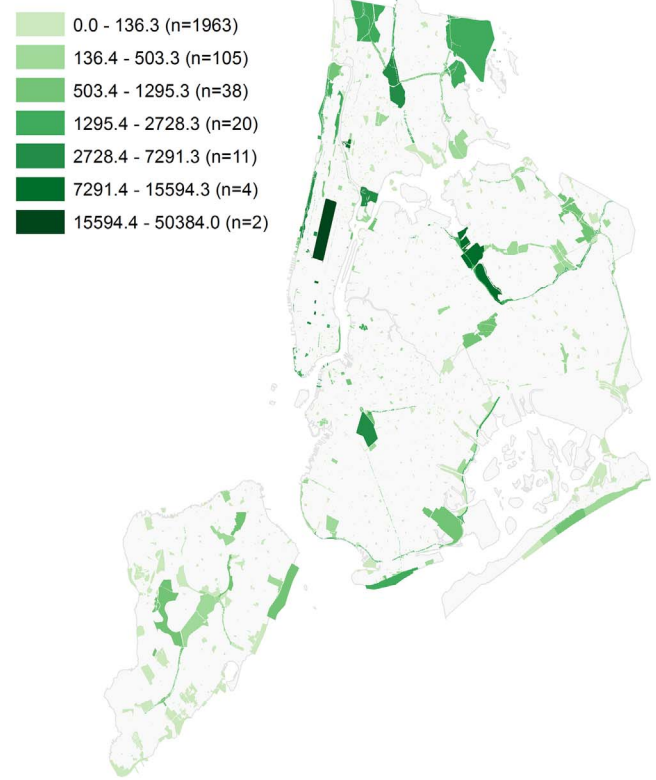


Fig. 2. Geographic distribution of Flickr photo user days (FUD) and Twitter user days (TUD). FUD and TUD are classified using Jenks optimization in order to illustrate variation given the positively-skewed distribution.

Table 2
Ranking of the ten most visited parks.

Rank	Flickr user days visitation	Mean FUD (2005–2014)	Twitter user days visitation	Mean TUD (2012–2014)
1	Central Park, Manhattan	5271.3	Central Park, Manhattan	50,384.0
2	Bryant Park, Manhattan	893.8	Flushing Meadows Corona Park, Queens	15,594.3
3	Union Square Park, Manhattan	729.2	Macombs Dam Park, Bronx	13,162
4	Flushing Meadows Corona Park, Queens	654.7	Bryant Park, Manhattan	12,354.3
5	Prospect Park, Brooklyn	645.9	Union Square Park, Manhattan	11,040.3
6	Washington Square Park, Manhattan	610.4	Washington Square Park, Manhattan	7291.3
7	Battery Park, Manhattan	594.7	Madison Square Park, Manhattan	6303.0
8	Brooklyn Bridge Park, Brooklyn	536.8	Theodore Roosevelt Park, Manhattan	5900.7
9	Coney Island Beach & Boardwalk, Brooklyn	486.8	Prospect Park, Brooklyn	5395.7
10	Bronx Park, Bronx	453.8	Lincoln Center Plaza, Manhattan	5017.3

media visitation rates occur in parks with more and larger proportions of water bodies, wifi hotspots, parks with more play area for children, in neighborhoods with higher proportions of White people (negatively correlated with Black or African American people) and near large roads. By contrast, flagship parks and parks in relatively high income neighborhoods are more highly used by tourists. Similar to the analyses that included residents, tourists and people who did not report their residential locations, the presence of a wifi hotspot, community park type, and proximity to bike routes and bus stops were important in both resident and tourist models (Table 4).

Whether or not a park contains data from Flickr or Twitter depends on a somewhat different set of factors (Table 5). Area of green space, hard courts, play area, playground park type, community park type and nature area park type were not important predictors of the occurrence of FUD and TUD in the logistic regression analyses. However, hard courts (which are correlated with soft courts), neighborhood-level population density, and neighborhood-level household income were all

positive predictors of the presence of Flickr photographs, though these variables were not as important in the model predicting variation in non-zero FUD alone (Table 3). Moreover, the final logistic regression model for TUD ($R^2 = 0.42$) was simplified to seven variables, including area (+), proximity to water bodies (+), population density (+), minority race (–), household income (+), proximity to nearest bus (+) and subway stop (+). Similar to the relatively large effect sizes noted above for the negative binomial regression analyses, the presence of a wifi hotspot increases the log-odds of FUD by 1.721, minority race decreases the log-odds of FUD and TUD by 1.956 and 0.727, respectively, and subway stops increase the log-odds of FUD by 0.200.

Models describing variability of FUD indicate that similar predictors explain the occurrence of FUD data for both residents and non-residents of New York City, as well as total FUD (Table 6). Logistic models for resident and tourist presence/absence of FUD indicate that flagship parks do not appear to drive visitation versus non-visitation, though neighborhood-level minority race (–), ethnicity (–) and vacancy (+)

Table 3

Multiple generalized negative binomial regressions for variation in non-zero Flickr photo user days (FUD) and Twitter user days (TUD).

	FUD non-zero N = 1234 R ² = 0.242				TUD non-zero N = 2001 R ² = 0.126			
	Estimate	Standard error	z	p > z	Estimate	Standard error	z	p > z
Intercept	2.673	0.159		< .001***	3.401	0.161	21.10	< .001***
Area (100 m ²)	4.71 × 10 ⁻⁵	1.74 × 10 ⁻⁵	2.71	< .01**	1.623 × 10 ⁻⁴	2.18 × 10 ⁻⁵	7.44	< .001***
Area grass/shrub and/or tree canopy (%)	-0.645	0.170	-3.80	< .001***	-0.616	0.120	-5.13	< .001***
Water bodies (count)	0.004	0.001	3.39	< .01**	0.003	0.001	2.74	< .01**
Distance to water bodies (100 m)	-0.106	0.009	-11.38	< .001***	-0.067	0.005	-13.05	< .001***
Hard courts: tennis, basketball, handball and netball (count)					0.060	0.009	6.94	< .001***
Wifi hotspot (presence)	1.220	0.198	6.16	< .001***	1.229	0.184	6.68	< .001***
Play area (100 m ²)	2.106 × 10 ⁻⁴	8.58 × 10 ⁻⁵	2.46	< .05*				
Playground (binary)	-1.184	0.124	-9.55	< .001***	-0.598	0.074	-8.09	< .001***
Community (binary)	1.590	0.154	10.33	< .001***	1.239	0.139	8.90	< .001***
Nature area (binary)					0.323	0.165	1.96	.05*
People/1000 m ²					0.017	0.003	5.20	< .001***
African American or Black people (% of total population)	-1.821	0.208	-8.77	< .001***	-0.864	0.117	-7.40	< .001***
Hispanic or Latino people (% of total population)	-2.395	0.236	-10.30	< .001				
Vacant units (% of total residential units)					3.523	0.786	4.49	< .001***
Median property value (\$1000)					3.751 × 10 ⁻⁴	1.03 × 10 ⁻⁴	3.64	< .001***
Length of bike routes (100 m)	6.015 × 10 ⁻⁴	2.991 × 10 ⁻⁴	2.01	< .05*	5.447 × 10 ⁻⁴	2.48 × 10 ⁻⁴	2.20	< .05*
Distance to nearest bike route (100 m)	-0.0717	0.015	-4.65	< .001***	-0.044	0.008	-5.76	< .001***
Distance to nearest bus stop (100 m)	-0.032	0.005	-5.83	< .001***	-0.0279	0.003	-8.76	< .001***
Subway stops (count)	0.447	0.031	14.28	< .001***	0.425	0.025	16.66	< .001***
Distance to nearest subway stop (100 m)	-0.012	0.002	-5.97	< .001***	-0.005	0.001	-3.57	< .001***
Interstate and state-recognized roads (count)	-0.102	0.032	-3.14	< .01**	0.071	0.026	2.79	< .01**
Dispersion parameter	1.489	0.072	16.83	< .001***	1.546	0.044	7.44	< .001***

Dispersion parameter significantly greater than zero indicates that the data are over dispersed and better estimated by a negative binomial than Poisson model.

*p < .05; **p < .01; ***p < .001.

Table 4

Multiple generalized negative binomial regressions for variation in non-zero resident Flickr photo user days (FUD) and tourist Flickr FUD.

	Non-zero resident FUD N = 902 R ² = 0.210				Non-zero tourist FUD N = 827 R ² = 0.246			
	Estimate	Standard error	z	p > z	Estimate	Standard error	z	p > z
Intercept	-0.2114	0.188	-1.82	.069	-0.895	0.141	-6.33	< .001***
Area buildings, pavement and/or road (%)	0.474	0.135	3.52	< .001***	0.985	0.164	5.99	< .001***
Area water (%)	1.652	0.683	2.42	< .05*				< .001***
Water bodies (count)	0.002	6.74 × 10 ⁻⁴	2.25	< .05*				< .01**
Distance to water bodies (100 m)	-0.043	0.007	-5.70	< .001***	-0.063	0.009	-6.84	< .001***
Wifi hotspot (presence)	0.975	0.133	7.35	< .001***	0.943	0.164	5.75	< .001***
Play area (100 m ²)	9.72 × 10 ⁻⁵	5.11 × 10 ⁻⁵	1.90	.057				
Playground (binary)	-0.501	0.103	-4.89	< .001***	-0.541	0.124	-4.34	< .001***
Flagship (binary)	0.856	0.270	3.18	< .01**	1.646	0.247	6.67	< .001***
Community (binary)	0.987	0.110	8.97	< .001***	1.085	0.132	8.23	< .001***
White people (% of total population)	1.120	0.146	7.66	< .001***				< .001***
Median household income (\$1000)					0.016	.002	10.04	< .001***
Length of bike routes (100 m)	6.285 × 10 ⁻⁴	2.079 × 10 ⁻⁴	3.02	< .01**	.001	2.555 × 10 ⁻⁴	4.57	< .001***
Subway stops (count)	0.274	0.020	13.25	< .001***	.293	.025	11.59	< .001***
Distance to nearest subway stop (100 m)	-0.005	0.023	-3.23	.001**	-.007	.002	-3.17	< .01**
Interstate and state-recognized roads (count)	-0.055	0.238	-2.32	< .05*				< .01**
Dispersion parameter	0.549	0.072		< .001***	0.782	0.052		< .001***

Dispersion parameter significantly greater than zero indicates that the data are over dispersed and better estimated by a negative binomial than Poisson model.

*p < .05; **p < .01; ***p < .001.

do. Community parks (+) and nature areas (−) are important drivers of whether or not residents visit parks (but not tourists), while vacant units (+) and proximity to nearest bus stop (+) are important predictors of whether tourists visit parks (but not residents).

4.4. Validation for nature areas

Results of a comparison between GSM visitation estimates and on-site visitor counts from 33 parks with nature areas show that FUD and TUD are both positively and significantly ($r = 0.575$ and $r = 0.759$, respectively; $p < .001$) correlated with empirical visitation rates. Twitter-based visitation rates match visitor counts more closely than Flickr-based visitation rates (Fig. 3).

5. Discussion

This study reveals that several aspects of park characteristics, neighborhood socio-demographics and accessibility are important determinants of park use based on visitation data from two social media platforms. Specifically, the area of the park, park type and facilities such as athletic courts, wifi and play areas drive visitation. These findings could be interpreted in multiple ways.

While the proportion of play area has a positive relationship with visitation, parks that are classified as playgrounds get fewer visitors. This might be explained by limited facilities and the rules regarding playground use. While play areas can be included in any park type, parks defined as playgrounds are small, standalone features in which play area usually makes up at least 50% of the total area (NYC

Table 5

Multiple logistic regressions for presence/absence of Flickr photo user days (FUD) and Twitter user days (TUD).

	FUD N = 2217 R ² = 0.27				TUD N = 1933 R ² = 0.42			
	Estimate	Standard error	z	p > z	Estimate	Standard error	z	p > z
Intercept	0.267	0.399	0.67	.504	−0.523	0.566	−0.92	.355
Area (100 m ²)	4.74×10^{-3}	7.73×10^{-4}	6.14	< .001***	0.118	0.014	8.16	< .001***
Water bodies (count)	0.018	0.008	2.35	< .05*				
Distance to water bodies (100 m)	−0.053	0.009	−5.70	< .001***	−0.070	0.016	−4.36	< .001***
Soft courts: soccer, baseball, softball, cricket, football, rugby (count)	0.907	0.220	4.12	< .001***				
Wifi hotspot (presence)	1.759	0.809	2.17	< .05*				
People/1000 m ²	0.024	0.007	3.43	.001**	0.107	0.018	6.07	< .001***
African American or Black people (% of total population)	−1.956	0.229	−8.52	< .001***	−0.727	0.407	−1.79	.074
Hispanic or Latino people (% of total population)	−1.554	−0.341	−4.56	< .001***				
Household income (\$1000)	0.010	0.004	2.41	< .05*	0.028	0.008	3.40	.001**
Vacant units (% of total residential units)	2.943	1.528	1.93	.05*				
Length of bike routes (100 m)	0.003	6.50×10^{-4}	4.17	< .001***				
Distance to nearest bike route (100 m)	−0.053	0.015	−3.67	< .001***				
Distance to nearest bus stop (100 m)	−0.021	0.006	−3.63	< .001***	−0.046	0.010	−4.54	< .001***
Subway stops (count)	0.200	0.067	2.98	< .01**				
Distance to nearest subway stop (100 m)	−0.014	0.024	−5.64	< .001***	−0.024	0.005	−5.06	< .001***

*p < .05; **p < .01; ***p < .001.

Department of Parks and Recreation, 2016). NYC regulations ban adults who are unaccompanied by children in playgrounds, thus rendering these parks inaccessible to many adults. In addition, playground users may lack mobile phones and therefore be undercounted by social media-based counts compared with traditional field-based survey counts.

Wifi is a relatively consistent predictor of visitation. This may indicate that the availability of free internet access encourages people to visit parks. Alternatively, wifi may be enabling people to share more social media, thus driving-up our social-media-based estimates of park use. More social media activity is likely where wifi is publically provided, and we are not able to ascertain from this study the extent to which that amenity is specifically sought out, or whether it leads to an overestimation of park visitors who would not otherwise post to social media.

Parks with greater areas of green space get fewer visits. Similarly, parks that are designated nature areas are less popular with residents (according to FUD), but more popular destinations among all visitors (both tourists and residents, according to TUD). These results are contrary to our expectations and appear to contradict previous studies showing the popularity of parks for nature-based recreation. However,

this may be because the green space variable in this study included manicured lawns, sports fields and highly landscaped areas that differ considerably in form from the richer natural habitats that cover some sites. The nature areas in NYC do contain more natural features such as marshland; however, they are generally vacant or unimproved areas that lack the trails and built features that would allow visitors to move easily throughout (NYC Department of Parks and Recreation, 2016). Even if parks with high coverage of green spaces and natural habitats are not the most highly visited urban areas, people who visit these parks derive physical and psychological health benefits (Tzoulas et al., 2007), and nature in urban parks may provide important ecosystem functions – such as air purification and climate moderation – that directly benefit people who live near parks that are actually green.

Previous studies have found that minority populations tend to have less access to parks, partially due to economic barriers and time constraints, and in part due to experiences of feeling unwelcome, unsafe and discriminated against (Byrne, 2012; Cutts et al., 2009; Floyd, 1999; Owens, 2002; Solop et al., 2003). In addition to these potential barriers, parks in neighborhoods with higher proportions of minorities may be smaller in size and have lower quality facilities (Crawford et al., 2008). A NYC-based study by Miyake et al. (2010) found that although park

Table 6

Multiple logistic regressions for presence/absence of resident Flickr photo user days (FUD) and tourist Flickr FUD.

	Resident FUD N = 2127 R ² = 0.29				Tourist FUD N = 2134 R ² = 0.30			
	Estimate	Standard error	z	p > z	Estimate	Standard error	z	p > z
Intercept	0.693	0.170	4.08	< .001***	0.321	0.238		.177
Area (100 m ²)	0.002	3.23×10^{-4}	5.87	< .001***	6.59×10^{-4}	1.24×10^{-4}	5.30	< .001***
Water bodies (count)					0.016	0.004	3.91	< .001***
Distance to water bodies (100 m)	−0.051	0.010	−5.32	< .001***	−0.062	0.011	−5.83	< .001***
Hard courts: tennis, basketball, handball and netball (count)	0.103	0.020	5.13	< .001***				
Wifi hotspot (presence)	1.479	0.542	2.73	< .01**				
Play area (100 m ²)				***	1.33×10^{-3}	2.05×10^{-4}	6.46	< .001***
Playground (binary)	−0.388	0.127	−3.05	< .01**	−0.604	0.135	−4.46	< .001***
Community (binary)	0.968	0.342	2.83	< .01**				
Nature area (binary)	−1.125	0.458	−2.46	< .05*				
African American or Black people (% of total population)	−2.519	0.228	−11.04	< .001***	−2.161	0.237	−9.14	< .001***
Hispanic or Latino people (% of total population)	−1.961	0.262	−7.47	< .001***	−2.107	0.289	−7.29	< .001***
Vacant units (% of total residential units)					3.928	1.505	2.61	< .01**
Length of bike routes (100 m)	0.003	5.11×10^{-4}	5.51	< .001***	.002	4.93×10^{-4}	4.78	< .001***
Distance to nearest bike route (100 m)	−0.060	0.016	−3.70	< .001***	−0.069	0.018	−3.82	< .001***
Distance to nearest bus stop (100 m)					−0.045	0.009	−5.10	< .001***
Subway stops (count)	0.383	0.061	6.28	< .001***	0.356	0.060	5.94	< .001***
Distance to nearest subway stop (100 m)	−0.012	0.003	−4.45	< .001***	−0.016	0.003	−5.20	< .001***

*p < .05; **p < .01; ***p < .001.

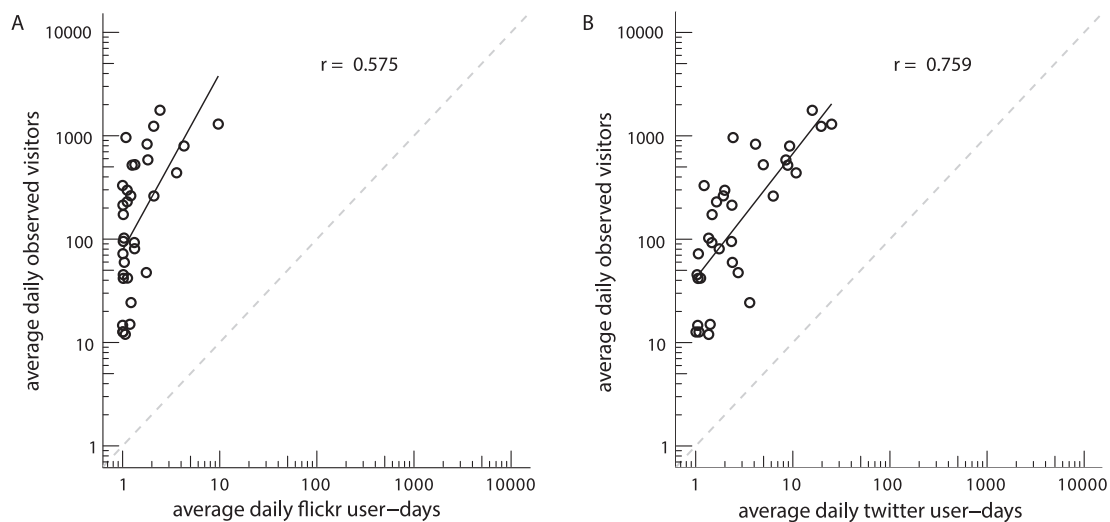


Fig. 3. Correlations in park nature areas between average daily Flickr user-days and average daily observed visitors (A) and between average daily Twitter user-days and average daily observed visitors (B).

access appears to be equitable based on the proportion of racial and ethnic minorities within 400 m and 800 m of parks, non-Hispanic White residents are more likely to live near the city's large parks, while Asian and Latino populations are more likely to live near small parks. The authors suggest that small parks may lack enough space needed for physical activities and facilities that mediate physical activity, which can be problematic from a public health equity perspective. Small park size and poor quality may explain our findings that minority proportion in the neighborhood surrounding the park has a substantial effect on social media-based visitation.

Nearly all of the variables measuring accessibility are important predictors of park visitation in our models – particularly the number of subway stops near parks. However, contrary to our hypothesis, neighborhood crime does not have a significant influence on visitation. This may be because popular and frequently visited parks and areas around frequently visited parks are being targeted for crime such as theft or with differences in patterns of reported versus actual crime. More research is needed to understand whether criminal activity is not actually a deterrent of park visitation.

We find minor differences in the factors associated with visitation according to the two social media platforms used to monitor visitors. Examples include hard courts, population density, vacant units and median property value, which influence TUD, but not FUD; play area and Hispanic proportion influence FUD but are not as important for TUD. Minority race and ethnicity have a smaller effect size (–) for TUD than FUD. To the extent that this may suggest Whites are less likely to frequent parks in neighborhoods with relatively high proportions of Blacks and Hispanics, this could imply that Twitter users are more racially and ethnically diverse than Flickr users. While there are strong similarities among factors that influence visitation by both residents and tourists, resident visitation is more strongly driven by community Parks (+) and nature areas (–), while tourist visitation is driven by vacant units (+) and bus stops (+), implying that tourists are more likely to use bus as a form of transportation for visiting parks.

This study was limited in that we did not measure park quality, including the quality of park facilities, park conditions or maintenance. These factors may be driving other proximate influences such as neighborhood race and ethnicity. In addition, many of the independent variables that we used to construct the predictive models are correlated with one another, making it difficult to ascertain exact drivers of visitation. For instance, the proportion of green space in parks is negatively correlated with the proportion of impervious surface. Relatively high proportions of impervious surface could indicate the presence of

facilities that draw people to parks. Although the proportion of impervious surface was not correlated with the facility types that we measured (including soft and hard athletic courts), other facilities that we were not able to measure (such as paved walkways) may be related to higher proportions of impervious surface. Number of facilities, park area and flagship parks are also correlated with one another. From a park management perspective, it is unclear whether increasing park sizes, the number of facilities in parks or park advertising would enhance visitation rates. Number of bus stops and subway stops are correlated with one another, making it difficult to ascertain whether one transit type is more effective at transporting visitors than another. Finally, White and Black populations are negatively correlated with one another. Correlated variables (≥ 0.5) were removed from regression models by selecting the variable with the highest p-value, but this removal could be masking influences that are not only redundant but also potentially additive.

This study uses data from social media platforms to estimate visitation rates, and therefore, our findings are not necessarily generalizable to the population of park users as a whole. On-site counts from a subset of parks in NYC suggest that Flickr and Twitter postings are reliable indicators of visitation (approximately 58–76% correlated with a small sample of on-site visitor counts) as concluded by other studies (e.g., Dunkel, 2015; Keeler et al., 2015; Sessions et al., 2016; Wood et al., 2013). The unexplained variability in this relationship might be due to biases in the social media data and to uncertainty in the observed visitation rates. Park use varies widely from day to day, and the relatively few observations from each park (approximately 90 min each on 3–8 days during the summer) highlight the challenge in accurately estimating daily use-rates based on observations, across a large network of parks. In addition, the residuals in these relationships suggest social media posting behavior varies by park. In this sample, less visited parks tend to have higher than expected FUD and TUD values, and more frequently visited parks have lower than expected social media values. Parks with relatively low visitation that is overestimated by social media values include Dubos Park, a wildlife sanctuary in Jamaica Bay, Idlewild Park, a public marsh near John F. Kennedy Airport and several shorefront parks in the Bronx including Riverdale and Spuyten Duyvil. Parks with relatively high visitation that is underestimated by social media values include several parks which have sports facilities such as Canarsie Park, Bayswater and Cunningham Park, as well as the waterfront park Bayswater. Underlying park-to-park differences in the demographics of visitors could be compounded by differences in who uses the social media platforms. Social media users are more likely to be

younger, more urban, and female compared to the larger U.S. population. Yet, use across platforms is relatively similar among Whites, Blacks, and Hispanics, as well as relatively similar across educational attainment and household income brackets, with a slight bias toward lower income brackets (Zickuhr, 2013). In addition, likelihood of social media use may vary by park type in some cases. For instance, young children who lack mobile phones may represent a relatively large user population of playgrounds, which would lead to under-counts of such park types. Issues of bias due to park type and park characteristics that encourage social media use or draw social media users in particular could be addressed through content analysis and field studies that examine finer-scale variability in park use drivers, as well as variation in type and intensity of use.

Further research is needed to ensure that TUD and FUD, or other social media-based indicators, provide representative data on actual park visitors with respect to both quantity and demographic makeup, and across the full range of urban parks. Visitation surveys using traditional count methods that stratify across park types and neighborhood demographics could be used to validate such indicators. In addition, further research is needed to understand the causal drivers of park visitation in ways that directly drive management decisions. In particular, quality measures would help to supplement information about recreational facilities and potentially explain variation in use across neighborhoods.

6. Conclusions

As access to high quality city parks and other public spaces becomes more crucial with increasing urban populations, visitation information will also become increasingly important for park management. Given that local governments often lack resources for intensive survey implementation, social media may offer supplementary data or even alternative ways to measure visitation on a regular basis. Here, we used Twitter and Flickr posts as data to examine variation across New York City's 2143 parks and correlates of variation in visitation. Similar to previous studies, findings suggest that park size, facilities, neighborhood socio-demographics and public transportation access are important determinants of visitation. However, our findings that green space is not a positive predictor – and in some cases may be a negative predictor – of park use does not conform with other studies about nature-based recreation, implying that although people derive benefits from nature, they may not be motivated to visit a park for the types of green space that are present. We also found that crime was not a significant predictor of park use, which may be related to heavy crime and/or heavy policing in high density neighborhoods with relatively high pedestrian activity. Further research is needed to validate social

media data as a park visitation indicator in the urban context by collecting traditional survey-based count data across a wide variety of park types. We suggest that such data, coupled with other spatially-explicit information about park qualities and the neighborhoods around parks could inform park managers of how to improve parks in ways that equitably distribute public resources across socio-demographic and socio-economic groups that have been historically underserved.

Through programs such as the Community Parks Initiative (City of New York, 2014), NYC and other urban communities are working to develop more equitable and sustainable park systems. We suggest that analyses such as our which examine use variation across a wide range of parks could support such programs by indicating which facilities should be part of park retrofit investments to encourage wider park use, particularly in minority communities that may have poor park access. While recreation is still a major driver of park visitation, wifi and other amenities may be changing uses of parks and require new kinds of analyses and investment to meet changing needs. Given the increasing ubiquity of GSM, cities may benefit from investing in new information infrastructures (e.g., integrating multiple GSM data streams, developing new apps, and tapping into citizen science) that allow for new ways of social sensing and interaction between city agencies and parks visitors. For instance, a unified space such as an interactive online map where information from multiple social media are overlaid can help make progress toward a more systematic representation of park visitation through social media and allow for public queries of the data at different points of time and in different urban communities. While more in-depth research is needed to validate GSM for park visitation, as social media users become more numerous and diverse, such data hold promise for generating new knowledge about who uses urban parks and why.

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Appendix A

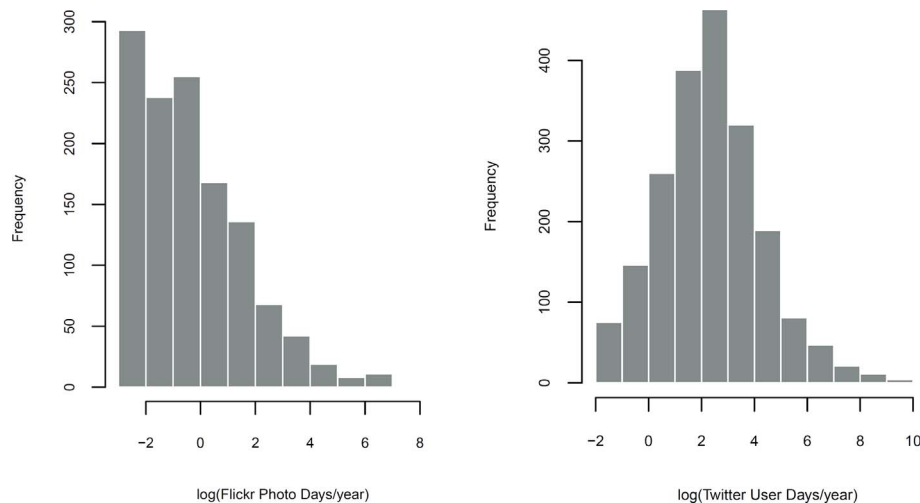


Fig. A1. Histograms of dependent variables – Flickr photo days/year (2005–2014) and Twitter user days/year (2012–2014) – showing skewed distributions.

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